

Practical No. 04

Connecting Two Networks with a Router and Two Switches, on Cisco Packet Tracer

Easiest and complete revision notes, written for someone starting with zero networking knowledge

What This Practical Is About, in One Paragraph

You build two small networks, each made of its own switch with two PCs plugged into it, and you join both of these networks together using a single router in the middle. Switch0's network uses the address range 192.168.1.x and Switch1's network uses the address range 192.168.2.x. PC1 and PC2, on Switch0, can already talk to each other immediately, since they share the same switch and the same network. The same is true for PC3 and PC4 on Switch1. But for a PC on Switch0's side to reach a PC on Switch1's side, the traffic has to leave its own network entirely, and that is only possible because the router sits in between with one leg plugged into each switch. Once the router's two interfaces are given IP addresses and switched on, the router already knows how to move traffic between the two networks, with no extra routing command needed at all, and that one idea is the whole point of this practical.

Key Terms Explained First

Router. A device that connects two or more different networks together and forwards traffic between them. Unlike a switch, which only moves data around inside one network, a router's whole job is moving data between separate networks.

Interface. A physical connection point on a router or switch where a cable plugs in, such as GigabitEthernet0/0. Each interface on a router usually belongs to a different network and gets its own IP address.

Directly connected network. A network that is plugged straight into one of the router's own interfaces. A router automatically knows how to reach every directly connected network the moment that interface is given an IP address and turned on, without you having to type any separate routing command.

Routing table. A list stored inside the router of every network it knows how to reach, and which interface to send traffic out of to get there. For this practical, the routing table fills itself in automatically because both networks are directly connected.

Default gateway. The IP address a PC sends its traffic to whenever the destination is outside its own network. For a PC on Switch0's side, this is the router's interface facing Switch0. For a PC on Switch1's side, it is the router's interface facing Switch1.

VLAN 1. The one default VLAN that already exists on every Cisco switch out of the box. Every switch port belongs to VLAN 1 unless it is moved elsewhere. This practical does not create any new VLANs, it simply uses the switch's own default VLAN 1 for everything.

SVI, Switch Virtual Interface. A logical, software based interface on a switch, written as `interface vlan 1` in the CLI, that lets you give the switch itself an IP address. A switch does not need this address to move data between its ports, it only needs it so that a person can remotely reach and manage the switch, for example over Telnet.

ip default-gateway. A command typed on the switch, telling the switch itself where to send its own management traffic if the destination is outside its own network. This is separate from anything to do with the PCs, it only affects the switch's own management access.

line vty 0 4. A command that selects all five virtual terminal lines used for remote access such as Telnet, so that a password can be applied to them. Without a password set here, a remote login attempt is normally refused.

enable password. A password that protects privileged mode, the mode that allows deeper configuration access on a device. Anyone trying to move from user mode into privileged mode remotely will be asked for this password.

Running config and startup config. Running config is the configuration currently active in the device's memory right now. Startup config is the version saved permanently to storage, loaded automatically every time the device restarts. Saving means copying the running config into the startup config.

write memory. A shorter, older command that does exactly the same job as `copy running-config startup-config`, it saves the current configuration permanently so it survives a restart.

Ping. A simple test message sent from one device to another to check whether it can be reached. Used here as the final proof that the two networks can talk to each other through the router.

Where Exactly Is the CLI, Step by Step

This is the same for the router and for both switches, so it only needs explaining once.

1. On the Cisco Packet Tracer workspace, left click once on the device icon, the router or either switch.
2. A small window pops up. At the top you will see a few tabs, usually named Physical, Config, and CLI.
3. Click on the tab named CLI.
4. A black terminal style screen appears, this is where every command below is typed.
5. If the screen shows a message asking about the initial configuration dialog, type `no` and press Enter, then press Enter again to get a clean prompt.

As before, type each command one line at a time and press Enter after every single line, rather than pasting a large block together.

The Topology You Are Building

One router in the middle, with one leg going to each of two switches, and two PCs hanging off each switch.

- PC1 and PC2 connect to Switch0, on ports FastEthernet0/1 and FastEthernet0/2.
- PC3 and PC4 connect to Switch1, on ports FastEthernet0/1 and FastEthernet0/2.
- Switch0 connects to the router's GigabitEthernet0/0 port, using Switch0's GigabitEthernet0/1 port.
- Switch1 connects to the router's GigabitEthernet0/1 port, using Switch1's GigabitEthernet0/1 port.
- All the ports involved here, on both switches and on the router, are plain access style connections. There is no trunk port and no VLAN tagging anywhere in this practical, since only VLAN 1 is used throughout.

The Full IP Address Plan

Device	Belongs To	IP Address	Subnet Mask	Default Gateway
PC1	Switch0 network	192.168.1.10	255.255.255.0	192.168.1.1
PC2	Switch0 network	192.168.1.11	255.255.255.0	192.168.1.1
PC3	Switch1 network	192.168.2.10	255.255.255.0	192.168.2.1
PC4	Switch1 network	192.168.2.11	255.255.255.0	192.168.2.1
Router Gi0/0	Switch0 network, gateway	192.168.1.1	255.255.255.0	not applicable
Router Gi0/1	Switch1 network, gateway	192.168.2.1	255.255.255.0	not applicable
Switch0 VLAN 1, management	Switch0 network	192.168.1.2	255.255.255.0	192.168.1.1
Switch1 VLAN 1, management	Switch1 network	192.168.2.2	255.255.255.0	192.168.2.1

Notice, exactly like the router-on-a-stick practical, that the router's interface address on each side is the exact same address every PC on that side uses as its default gateway. The router interface literally is the gateway, on both sides this time, instead of through subinterfaces.

Step 1, Assign IP Address and Default Gateway on Each PC

PC1, PC2, PC3, PC4 — Desktop

1. Click on PC1, go to Desktop, then IP Configuration.
2. Enter IP Address 192.168.1.10, Subnet Mask 255.255.255.0, and Default Gateway 192.168.1.1.
3. Repeat this for PC2, PC3, and PC4 using the exact values from the table above. This part does not use CLI at all, it is done through the graphical Desktop screen, exactly like earlier practicals.

Step 2, Configure Router0, with Every Command Explained

ROUTER0 — CLI

Command	What It Actually Does
enable	Moves you from basic user mode into privileged mode, allowing deeper access to the device.
configure terminal	Enters global configuration mode, where device wide settings are changed.
interface gigabitEthernet 0/0	Selects the router's first physical port, the one wired to Switch0, so the next commands only apply to this one interface.
ip address 192.168.1.1 255.255.255.0	Gives this interface the address that every PC on Switch0's side will use as its default gateway.
no shutdown	Router interfaces are turned off by default, this command turns the port on. Forgetting this on either interface is one of the most common reasons the whole lab fails.
exit	Leaves interface configuration mode and returns to global configuration mode.
interface gigabitEthernet 0/1	Selects the router's second physical port, the one wired to Switch1.
ip address 192.168.2.1 255.255.255.0	Gives this interface the address that every PC on Switch1's side will use as its default gateway.
no shutdown	Turns this second interface on as well, exactly as with the first one.
exit	Leaves interface configuration mode again.
line vty 0 4	Selects all five virtual terminal lines, used for remote access such as Telnet, so a password can be set on them.
password cisco	Sets the password that will be asked for on any remote Telnet connection to this router.
login	Tells the router to actually require that password before letting a remote session in, without this the password alone is not enforced.
exit	Leaves line configuration mode, back to global configuration mode.
enable password cisco	Sets the password needed to move into privileged mode, protecting deeper configuration access.
write memory	Saves the current configuration permanently, so it survives a restart. This is the same result as <code>copy</code>

	running-config startup-config , just typed as a shorter, older command.
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The Telnet and enable password lines are optional, marked so in the lab, and are only needed if you plan to manage the router remotely rather than through the CLI tab directly. The two `interface` blocks with `ip address` and `no shutdown` are the only commands that are actually required for the two networks to reach each other.

Step 3, Configure Switch0, with Every Command Explained

SWITCH0 — CLI, connected to PC1, PC2, and Router Gi0/0

Command	What It Actually Does
<code>enable</code>	Enters privileged mode on the switch.
<code>configure terminal</code>	Enters global configuration mode on the switch.
<code>hostname Switch0</code>	Renames the device to Switch0 in the CLI prompt, purely so you can tell it apart from Switch1, this step is optional.
<code>interface vlan 1</code>	Enters configuration mode for the switch's own management interface, the default VLAN 1 SVI.
<code>ip address 192.168.1.2 255.255.255.0</code>	Gives the switch itself a management address on the same network as PC1 and PC2, so it can be reached and managed remotely.
<code>no shutdown</code>	Activates this management interface, without it the switch's own IP address will not respond even though the switch keeps forwarding PC traffic normally.
<code>exit</code>	Leaves interface configuration mode.
<code>ip default-gateway 192.168.1.1</code>	Tells the switch where to send its own management traffic if a request comes from outside its own network, this only affects the switch, not the PCs.
<code>interface fastEthernet 0/1</code>	Selects the port that PC1 is plugged into.
<code>switchport mode access</code>	Marks this port as an access port, connected to a single normal end device.
<code>switchport access vlan 1</code>	Places the port in VLAN 1, the switch's default VLAN, which is all that is needed here since no other VLANs exist in this practical.
<code>no shutdown</code>	Makes sure the port is active.
<code>exit</code>	Leaves interface configuration mode.
<code>interface fastEthernet 0/2</code>	Selects the port that PC2 is plugged into, and the same four lines below are applied to it.
<code>switchport mode access</code>	Marks Fa0/2 as an access port too.
<code>switchport access vlan 1</code>	Places Fa0/2 in VLAN 1 as well.
<code>no shutdown</code>	Activates Fa0/2.
<code>exit</code>	Leaves interface configuration mode.
<code>interface gigabitEthernet 0/1</code>	Selects the port wired across to the router's Gi0/0 interface.
<code>switchport mode access</code>	Since only one VLAN exists in this whole practical, this link to the router is kept as a simple access port rather than a trunk, unlike the router-on-a-stick practical.
<code>switchport access vlan 1</code>	Places this port in VLAN 1, matching every other port on this switch.
<code>no shutdown</code>	Activates the port facing the router.
<code>exit</code>	Leaves interface configuration mode.
<code>line vty 0 4</code>	Selects the virtual terminal lines for optional remote Telnet access to this switch.
<code>password cisco</code>	Sets the password required for a remote Telnet session.
<code>login</code>	Enforces that password on incoming remote sessions.
<code>exit</code>	Leaves line configuration mode.
<code>write memory</code>	Saves this configuration permanently on Switch0.

Step 4, Configure Switch1, with Every Command Explained

SWITCH1 — CLI, connected to PC3, PC4, and Router Gi0/1

Switch1 is configured exactly the same way as Switch0, only the network numbers change, from 192.168.1.x to 192.168.2.x.

Command	What It Actually Does
<code>enable</code>	Enters privileged mode on the switch.

configure terminal	Enters global configuration mode.
hostname Switch1	Renames the device to Switch1, optional, purely for clarity.
interface vlan 1	Enters configuration mode for the management SVI.
ip address 192.168.2.2 255.255.255.0	Gives Switch1 a management address on the same network as PC3 and PC4.
no shutdown	Activates the management interface.
exit	Leaves interface configuration mode.
ip default-gateway 192.168.2.1	Tells Switch1 where to send its own management traffic when the destination is outside 192.168.2.x.
interface fastEthernet 0/1	Selects the port that PC3 is plugged into.
switchport mode access	Marks the port as an access port.
switchport access vlan 1	Places Fa0/1 in VLAN 1.
no shutdown	Activates Fa0/1.
exit	Leaves interface configuration mode.
interface fastEthernet 0/2	Selects the port that PC4 is plugged into.
switchport mode access	Marks Fa0/2 as an access port.
switchport access vlan 1	Places Fa0/2 in VLAN 1.
no shutdown	Activates Fa0/2.
exit	Leaves interface configuration mode.
interface gigabitEthernet 0/1	Selects the port wired across to the router's Gi0/1 interface.
switchport mode access	Kept as a plain access port, same reasoning as on Switch0.
switchport access vlan 1	Places this port in VLAN 1.
no shutdown	Activates the port facing the router.
exit	Leaves interface configuration mode.
line vty 0 4	Selects the virtual terminal lines for optional remote Telnet access.
password cisco	Sets the remote login password.
login	Enforces that password.
exit	Leaves line configuration mode.
write memory	Saves this configuration permanently on Switch1.

Step 5, Verification Commands

ROUTER0 — CLI

Command	What It Actually Does
show ip interface brief	Lists every interface on the router along with its IP address and whether it is up, used to confirm both Gi0/0 and Gi0/1 show an address and a status of up.
show ip route	Displays the router's routing table, you should see both 192.168.1.0 and 192.168.2.0 listed automatically as directly connected networks, marked with a C.

SWITCH0 and SWITCH1 — CLI

Command	What It Actually Does
show vlan brief	Confirms every port landed in VLAN 1 as intended.
show ip interface brief	Confirms the VLAN 1 management interface shows the correct IP address and a status of up.

Step 6, Testing, in the Correct Order

Test 1, same switch, should work immediately. On PC1, go to Desktop, then Command Prompt, and type:

```
ping 192.168.1.11
```

This is pinging PC2, which shares the same switch and the same network as PC1. This succeeds even before the router is touched at all, since Switch0 alone can already deliver traffic between its own ports.

Test 2, across the router, different network. Still on PC1, type:

```
ping 192.168.2.10
```

This is pinging PC3, which sits on the completely different 192.168.2.x network behind Switch1. Once the router's two interfaces are correctly addressed and turned on, this succeeds too, because the router already has both networks in its routing table as directly connected, with no extra routing command required. The very first attempt may show Request timed out, this is normal, it is just ARP learning addresses for the first time, simply run the ping again.

Troubleshooting, if a Ping Still Fails

- Check that Switch0's Gi0/1 port really is the one wired to the router's Gi0/0 port, and Switch1's Gi0/1 to the router's Gi0/1, crossed cabling here is a common mistake.
- On the router, confirm `no shutdown` was typed under both `interface gigabitEthernet 0/0` and `interface gigabitEthernet 0/1`, a missed one leaves that whole side of the network unreachable.
- Confirm every PC has the correct default gateway matching its own side, a PC on Switch0's side must use 192.168.1.1, not 192.168.2.1.
- Confirm the two switches were not accidentally left with their router facing port set to trunk mode, this practical only needs access mode throughout, since it uses just VLAN 1.
- On the router, run `show ip route` and check that both 192.168.1.0 and 192.168.2.0 actually appear, if one is missing, its matching interface is either not addressed correctly or still administratively down.
- Give it a second try, the first ping after any fresh configuration change commonly times out once due to ARP, and then works normally.

Viva Questions for This Practical

Q1. What is the main job of a router?

To connect two or more separate networks together and forward traffic between them, unlike a switch, which only moves data within a single network.

Q2. Why did the router not need any separate routing command to reach both networks?

Because both networks are directly connected, each one plugged straight into one of the router's own interfaces, and a router automatically adds a directly connected network to its routing table the moment that interface is addressed and turned on.

Q3. What would happen if `no shutdown` was forgotten on one router interface?

That interface would stay administratively down, so its entire side of the network would be unreachable from the other side, even though every other setting was correct.

Q4. Why does the switch need an IP address at all, if it only works at Layer 2?

The switch's own IP address, set on the VLAN 1 SVI, is only used so the switch itself can be reached and managed remotely, it plays no part in how the switch actually forwards traffic between its ports.

Q5. What is the difference between this practical and the router-on-a-stick practical?

Here, two separate switches each carry only one VLAN, connected to two separate physical router interfaces. In router-on-a-stick, one switch carries two VLANs over a single trunk port, and the router splits one physical interface into two subinterfaces to tell them apart.

Q6. Why was the switch to router link kept as an access port instead of a trunk?

Because only one VLAN, VLAN 1, exists anywhere in this practical, so there is no need to tag or separate multiple VLANs over that link, an access port is enough.

Q7. What does `ip default-gateway` do on a switch?

It tells the switch where to send its own management traffic when the destination lies outside the switch's own network, it has no effect on how PC traffic is forwarded through the switch.

Q8. What command shows the router's routing table?

```
show ip route
```

Q9. What is the difference between `write memory` and `copy running-config startup-config`?

Nothing functionally, they do the exact same job of saving the current configuration permanently, `write memory` is just an older, shorter way of typing the same instruction.

Q10. Why did the ping to PC2 succeed before any router configuration was done at all?

Because PC1 and PC2 are on the same switch and the same network, a switch alone can already deliver traffic between its own ports without any router involved.

Q11. What does `line vty 0 4` followed by `login` actually achieve together?

`line vty 0 4` selects the five virtual terminal lines used for remote access, and `login` tells the device to actually demand the configured password on those lines before letting a remote session in.

Q12. If Switch0's management IP address stopped responding, would PC1 and PC2 still be able to reach PC3 and PC4?

Yes, the switch's own management IP address is completely separate from its job of forwarding PC traffic, so PC to PC connectivity through the switch and router would be unaffected.

Important Things to Remember

- A router automatically knows how to reach any network that is directly connected to one of its own interfaces, no static route or routing protocol is needed for that network to appear in the routing table.
- `no shutdown` must be typed on every router interface used in this practical, router interfaces are administratively down by default, unlike most switch ports.
- The switch's own IP address, on the VLAN 1 SVI, is only for managing the switch remotely, it is not required for PCs to communicate

through it.

- Since this practical only uses VLAN 1 throughout, every port involved, on both switches and towards the router, is a plain access port, there is no trunk port anywhere in this lab.
- Each PC's default gateway must match the router interface facing its own side, a PC on Switch0's side must never be given Switch1's gateway address.
- `write memory` and `copy running-config startup-config` save the configuration in exactly the same way, save your work on the router and on both switches so nothing is lost on restart.
- The very first ping after any fresh configuration change may time out once due to ARP still learning addresses, this is normal, simply try again.
- Type commands one line at a time instead of pasting large blocks, it avoids CLI errors and helps you actually remember what each line does for the viva.

Quick recall order: PC IP and gateway, Router0 both interfaces with no shutdown, Switch0 management IP and access ports, Switch1 management IP and access ports, save config on all three, then test same switch, then test across the router.